Worksheet 5.3 — Kinematics: Velocity & Acceleration

Grade 12 Calculus and Vectors (MCV4U) · Unit 5: Applications of Derivatives

The Big Ideas

$v = s'$, $a = v' = s''$, speed $= |v|$. At rest when $v = 0$; speeding up when v, a share a sign.

- $v(t) = s'(t)$, $a(t) = s''(t)$.
- At rest: $v = 0$; direction: sign of v.
- Speeding up: v, a same sign; slowing: opposite.

The Big Picture

Position, velocity, acceleration

If $s(t)$ is position, then velocity is $v(t) = s'(t)$ and acceleration is $a(t) = v'(t) = s''(t)$ each a derivative of the last.

Reading motion

Velocity's sign gives direction; where $v = 0$ the object is momentarily at rest and may turn around.

Speeding up or slowing

The object speeds up when velocity and acceleration share a sign, and slows when their signs differ.

Worked Examples

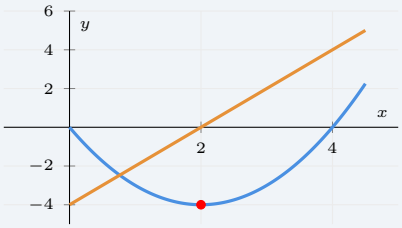
Example 1 — v and a

For $s(t) = t^2 - 4t$, find v and a.

Solution. **Step 1 (velocity is the derivative of position):** $v(t) = s'(t) = 2t - 4$.

Step 2 (acceleration is the derivative of velocity): $a(t) = v'(t) = 2$.

Conclusion: $v(t) = 2t - 4$, $a(t) = 2$. Velocity is the slope of position; at rest at the lowest point ($t = 2$):



Example 2 — At rest

When is $s(t) = t^2 - 4t$ at rest?

Solution. **Step 1 (find $v(t)$):** $v(t) = 2t - 4$.

Step 2 (set $v = 0$, since "at rest" means zero velocity): $2t - 4 = 0$.

Conclusion: $t = 2$ s.

Example 3 — Multiple rest times

When is $s(t) = t^3 - 6t^2 + 9t$ at rest?

Solution. **Step 1 (find $v(t)$):** $v(t) = 3t^2 - 12t + 9$.

Step 2 (set $v = 0$ and factor): $3(t^2 - 4t + 3) = 3(t-1)(t-3) = 0$.

Conclusion: at rest at $t = 1$ and $t = 3$.

Example 4 — Direction & accel

Describe the motion of $s(t) = t^3 - 6t^2 + 9t$ at $t = 0$.

Solution. **Step 1 (find $v(t)$ and evaluate at $t = 0$):** $v(t) = 3t^2 - 12t + 9 \Rightarrow v(0) = 9 > 0$, so the object is moving in the positive direction.

Step 2 (find $a(t)$ and evaluate at $t = 0$): $a(t) = 6t - 12 \Rightarrow a(0) = -12 < 0$.

Step 3 (interpret): positive velocity but negative acceleration means it's slowing down (decelerating) while still moving forward.

Conclusion: at $t = 0$ the object moves right ($v(0) = 9 > 0$) while decelerating ($a(0) = -12 < 0$).

Example 5 — Speeding up?

Is $s(t) = t^3 - 6t^2 + 9t$ speeding up at $t = 4$?

Solution. **Step 1 (evaluate $v(4)$):** $v(t) = 3t^2 - 12t + 9 \Rightarrow v(4) = 48 - 48 + 9 = 9 > 0$.

Step 2 (evaluate $a(4)$): $a(t) = 6t - 12 \Rightarrow a(4) = 24 - 12 = 12 > 0$.

Step 3 (apply the rule: same sign means speeding up): $v(4)$ and $a(4)$ are both positive.

Conclusion: yes, speeding up at $t = 4$.

Example 6 — Velocity

For $s(t) = t^2 - 6t$, find $v(t)$.

Solution. **Step 1 (differentiate $s(t)$):** $v(t) = s'(t)$.

Conclusion: $v(t) = 2t - 6$.

Example 7 — At rest

When is $s(t) = t^2 - 6t$ at rest?

Solution. **Step 1 (find $v(t)$):** $v(t) = 2t - 6$.

Step 2 (set $v = 0$ and solve): $2t - 6 = 0$.

Conclusion: $t = 3$.

Example 8 — Acceleration

For $s(t) = t^3 - 3t^2$, find $a(t)$.

Solution. **Step 1 (find $v(t)$):** $v(t) = 3t^2 - 6t$.

Step 2 (differentiate $v(t)$ to get $a(t)$): $a(t) = 6t - 6$.

Conclusion: $a(t) = 6t - 6$.

Example 9 — Max height

A projectile has $h(t) = -5t^2 + 20t$. When is it highest?

Solution. **Step 1 (find $v(t)$):** $v(t) = h'(t) = -10t + 20$.

Step 2 (set $v = 0$, since the height is maximized when the velocity is momentarily zero): $-10t + 20 = 0$.

Conclusion: $t = 2$ s.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

For $s(t) = t^2 - 6t$, find $v(t)$.

Question 2

When is $s(t) = t^2 - 6t$ at rest?

Question 3

For $s(t) = t^3 - 3t^2$, find $a(t)$.

Question 4

A projectile has $h(t) = -5t^2 + 30t$. When is it highest?

Question 5

If $v > 0$ and $a < 0$, is the object speeding up or slowing down?

Question 6

For $s(t) = 2t^2 - 8t$, find $v(t)$ and $a(t)$.

Question 7

When is $s(t) = t^3 - 6t^2 + 9t$ at rest?

Question 8

For $s(t) = t^3 - 3t^2$, when is the object at rest?

Question 9

At $t = 0$, $s(t) = t^3 - 6t^2 + 9t$: which direction is it moving?

Question 10

Speed of $s(t) = t^2 - 4t$ at $t = 0$?

Question 11

If $v < 0$ and $a < 0$, is the object speeding up or slowing down?

Question 12

Acceleration of $s(t) = t^2 - 4t$ (constant)?

Question 13 — Challenge

For $s(t) = t^3 - 6t^2 + 9t$, determine whether it is speeding up or slowing down at $t = 2$.

Question Recap & Answer Key

Question Recap

1. For $s(t) = t^2 - 6t$, find $v(t)$.
2. When is $s(t) = t^2 - 6t$ at rest?
3. For $s(t) = t^3 - 3t^2$, find $a(t)$.
4. A projectile has $h(t) = -5t^2 + 30t$. When is it highest?
5. If $v > 0$ and $a < 0$, is the object speeding up or slowing down?
6. For $s(t) = 2t^2 - 8t$, find $v(t)$ and $a(t)$.
7. When is $s(t) = t^3 - 6t^2 + 9t$ at rest?
8. For $s(t) = t^3 - 3t^2$, when is the object at rest?
9. At $t = 0$, $s(t) = t^3 - 6t^2 + 9t$: which direction is it moving?
10. Speed of $s(t) = t^2 - 4t$ at $t = 0$?
11. If $v < 0$ and $a < 0$, is the object speeding up or slowing down?
12. Acceleration of $s(t) = t^2 - 4t$ (constant)?
13. For $s(t) = t^3 - 6t^2 + 9t$, determine whether it is speeding up or slowing down at $t = 2$.

✓ Answer Key

1. $2t - 6$
2. $t = 3$
3. $a = 6t - 6$
4. $v = -10t + 30 = 0 \Rightarrow t = 3$ s
5. slowing down
6. $v = 4t - 8$, $a = 4$
7. $t = 1, 3$
8. $v = 3t^2 - 6t = 0 \Rightarrow t = 0, 2$
9. right ($v(0) = 9 > 0$)
10. $|v(0)| = |-4| = 4$
11. speeding up
12. 2
13. $v(2) = -3 < 0$, $a(2) = 0$ at an instant of zero acceleration (neither)